
Data Scientist

SUMMARY: Data Scientist with a clinical research and life sciences background: 10+ years in research, 6 years within hospital systems, and cross-industry consulting experience in between. Analyzes healthcare data to validate clinical tools and hospital programs, support executive decisions, and track community health trends.

TECHNICAL SKILLS

- Python, R, SQL
- Excel, Power BI, Tableau
- AWS (S3, EC2), Docker
- Epic Clarity, REDCap
- Scikit-learn, TensorFlow
- PyTorch, XGBoost, OpenCV, NLTK
- Pandas, NumPy, Matplotlib
- Git / GitHub, ArcGIS, Jupyter

HARD SKILLS

- Machine Learning / Deep Learning
- ML Validation / Bias Assessment
- Clinical Data Processing
- Generative AI / LLMs
- Cloud / Distributed Computing
- Statistics / Predictive Modeling
- Data Privacy / HIPAA Compliance
- Structured & Unstructured Data

SOFT SKILLS

- Cross-Disciplinary Collaboration
- Translating Technical Concepts
- Interpersonal Communication
- Creative Problem-Solving
- Managing Competing Priorities
- Bilingual Communication
- Self-Motivated Learner
- Curiosity for Discovery

PROFESSIONAL EXPERIENCE

Consultant

JUN 2020 – Present

Data Scientist / ML Engineer

- **Process Automation:** Wrote Python scripts to automate manual data workflows for clients, reducing errors and saving an average of ~\$5,000/week in labor and opportunity costs.
- **Machine Learning Development:** Built and deployed client-specific ML models (e.g., K-Means, XGBoost) for use cases such as customer segmentation and demand forecasting.
- **Cloud Migration:** Migrated clients to cloud infrastructure (AWS S3, EC2) to reduce costs and support computationally intensive workloads.
- **Data Dashboards:** Built interactive dashboards (Power BI, Excel) giving business leaders a clear, visual breakdown of company performance to support faster, more profitable decisions.
- **Computer Vision & Document Analysis:** Built OpenCV-based tools to scan 10TB+ databases and flag duplicate records with ~97% accuracy, helping clients quickly identify actionable opportunities.
- **Stakeholder Communication:** Served as the bridge between technical teams and business leadership, translating ML methodology into plain language clients could act on with confidence.
- **Data Governance:** Maintained strict data privacy and security standards while handling sensitive financial and PII datasets across multiple industries.

Trinity Health Hospital, Livonia, MI

JAN 2023 – JAN 2024

Senior Data Scientist

- **Clinical Model Evaluation:** Led the hospital's investigation into Epic's built-in predictive tools (e.g., the Sepsis Cognitive Model), reverse-engineering black-box algorithms and evaluating performance via AUC/ROC analysis to inform safe clinical screening protocols.
- **Data Pipelines & Algorithms:** Built automated data feeds (Python, R, SQL) powering live Tableau dashboards that measured whether the hospital's new virtual nursing initiative actually outperformed what it replaced.
- **Epic Clarity Queries:** Queried the Epic Clarity database, navigating complex data dictionaries to extract specific patient information needed to answer critical clinical questions.
- **Physician Collaboration:** Partnered with physicians and hospital leadership to build dashboards translating technical methodology into plain language for clinical and administrative staff.
- **Data Governance:** Maintained strict HIPAA compliance and rigorous data privacy standards while holding master access to sensitive patient health records.

Cottage Health Research Institute, Santa Barbara, CA

JUN 2019 – DEC 2022

Research Data Analyst

- **Predictive Modeling:** Developed boosted tree algorithms to model the relationship between patient data and C. difficile status, achieving a 5% improvement in prediction accuracy over the prior year's model.
- **Epidemiological Trend Tracking:** Built an ArcGIS-based hotspot mapping framework tracking C. difficile by acquisition type (community, hospital-acquired outpatient, hospital-acquired inpatient), then adapted it to track COVID-19 spread within the Santa Barbara community.
- **Data Pipelines:** Built reusable, time-saving data analysis pipelines covering over one million electronic health records.
- **Database Administration & Staff Training:** Served as REDCap database administrator for the hospital and led workshops to build staff data analysis and visualization skills.
- **Data Visualization:** Used Power BI, Excel, and Matplotlib to create visualizations presented to hospital executives, fellow researchers, and conference audiences nationwide.

PIH Health Hospital, Whittier, CA

OCT 2016 – JAN 2018

Research Data Coordinator

- **Clinical Trial Management:** Coordinated a multi-site clinical trial team (physicians, nurses, analysts, coordinators) through Phases II–III.
- **Predictive Modeling & Health Literacy:** Used EHR data and predictive modeling (e.g., Hidden Markov Models) alongside shared decision-making tools to promote health literacy among underrepresented patients managing obesity-related comorbidities like joint pain, hypertension, and type 2 diabetes.
- **Bilingual Data Collection:** Facilitated clean data collection by communicating with patients in English and Spanish, translating feedback into structured data in REDCap.

University of California, Riverside — Dept. of Psychology

SEP 2015 – JUN 2016

Research Assistant

- **Human Subjects Research:** Conducted human subjects research on iconic memory, working with 100+ participants over a 10-month period.
- **Quantitative Neurological Testing:** Administered computational neuroscience memory tests, EEG recordings, and eye-tracking experiments using MATLAB to collect quantitative data on human memory.

University of California, Riverside — Dept. of Botany & Plant Sciences

MAR 2014 – JUN 2015

Research Assistant

- **Statistical Analysis:** Performed statistical significance testing (e.g., ANOVA) to evaluate biofuel efficiency from transgenic tobacco plants.
- **Wet Lab Data Collection:** Completed 20+ wet lab procedures to prepare glycosyl composition analyses of tobacco cell walls and alcohol-insoluble residues.

EDUCATION

M.S. Analytics

Georgia Institute of Technology

B.A. Neuroscience

University of California, Riverside

LANGUAGE & CERTIFICATIONS

- Spanish (Read, Write, Speak – Fluent/Native)
- “Protecting Human Research Participants” – NIH
- “Essentials of Research Administration” – CITI

SELECTED PROJECTS

- **Genetic Disease Classification:** Built a machine learning model to classify and predict canine diseases from genetic composition data (M.S. Capstone).
- **COVID-19 Severity Modeling:** Developed a machine learning framework to predict severe COVID-19 disease outcomes.
- **NICU Quality Improvement:** Co-authored a peer-reviewed study (Neonatal Network, 2022) on developmental positioning in a Level III NICU, leading the data management strategy and statistical analysis (R, SciPy).
- **Opioid Crisis Visualization:** Built data visualizations analyzing trends in the U.S. opioid crisis.